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RESEARCH ARTICLE

Cryptanalysis of the Quantum Blind Signature Scheme for Supply Chain Finance and Its Improvement

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ABSTRACT

Recently, a quantum blind signature scheme based on the χ-state for quantum supply chain finance has been proposed, which utilizes blindness to protect the privacy of participating firms. However, our analysis found that this scheme cannot resist the intercept-measurement-resend attack and the entanglement-measurement attack, and the blind operation is ineffective against the insider attacker. Importantly, the blind signer can achieve existential forgery through a valid signature. Then, an improved scheme was proposed that reduces quantum bit consumption, eliminates the need for quantum state swap tests, and extends the message space for signatures to the more universal classical bit space. We have constructed a security model that formally defines the security properties and adversary capabilities of the quantum blind signature. Through this model, the blindness and unforgeability of the improved scheme are reduced to the principles of quantum mechanics in four games. The vulnerabilities of the original scheme and the feasibility of the improved scheme in quantum computing have been verified through the Qiskit simulator.

1 | Introduction

As the core technology to guarantee data integrity, authentication, and undeniability in the digital era, the digital signature is widely used in fields such as e-government, financial transactions. However, the emergence of quantum algorithms [1] has made it challenging to secure digital signatures that rely on mathematically difficult problems (such as large integer factorization and discrete logarithm problems).

The quantum signature (QS) [2, 3] achieves theoretical unconditional security because its security is built on the basic principles of quantum mechanics. Importantly, QSs achieve the same security properties as digital signatures and have been widely researched. In the field of QSs, the quantum blind signature (QBS) [4, 5] has received more attention due to its privacy-preserving properties. The QBS allows a blind signer to generate a valid signature without knowing the signature message, which is effective when deployed in scenarios where the privacy of the participants needs to be protected.

The QBS originally appeared in the scheme constructed by Wen et al. [6], and several schemes have been proposed since then. Zhu et al. [7] used the quantum Fourier transform to realize encryption and the protocol flow of a QBS through N-dimensional quantum states. Xia et al. [8] proposed an eavesdropping detection mechanism and utilized semi-quantum ideas to reduce the quantum cost of their QBS scheme. However, Cao et al. [5] proved thatthe scheme [8] fails to satisfy the unforgeability requirement and gave an improved version. In order to be applicable to voting and bidding scenarios, Qiang et al. [9] utilized the correlation of five-particle states to implement QBS schemes and auditing,

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where the security of the schemes is guaranteed by the quantum one-time pad (QOTP) and the BB84 protocol. Qiu et al. [10] proposed a bidding protocol based on the QBS and a two-state vector formalism, which can prevent foul play by malicious participants. In addition, QBS schemes without entangled states have been proposed to reduce the utilization of quantum resources. Zhong et al. [11] resolved the tension between anonymity and controllability by introducing negotiated public information and utilizing the QOTP to remove the need for entangled states. Xing et al. [12] constructed a QBS scheme with dense coding using single photons and Pauli gates and reduced the quantum cost using quadratic distributions and rearrangements. Although many QBS schemes have been proposed and some scenarios have been considered for their application, the potential of QBS in other scenarios remains unexplored.

Recently, Feng et al. [13] proposed a QBS scheme for supply chain finance (SCF-QBS) due to their consideration of scenarios combining quantum finance and quantum technology [14, 15]. In quantum supply chain finance, the main participants are core enterprises (CN), small and medium enterprises (SME), and financial institutions (FI). In the SCF-QBS scheme, the business request is treated as a QBS. The SME generates a signature to indicate the submission of information to The CN for credit rating and appraisal development. The FI is required to generate a blind signature to participate in the process, and The CN validates the signature for review. Feng et al. emphasized the need for SME privacy to be protected, which requires The FI to complete the signature without knowing business information of the SME. The SCF-QBS scheme seems to complete the required process.

However, we analyzed the SCF-QBS scheme and found that it cannot resist the intercept-measurement-resend (IMR) attack and the entanglement-measurement (EM) attack. Through these attacks, an internal adversary can easily break the protection of SMEs' privacy. Importantly, the SCF-QBS scheme cannot resist the existential forgery attack. In fact, many encryption algorithms in QS rely on combinations of quantum logic gate operations. Due to the unitary property of quantum logic gates, QS schemes are extremely vulnerable to the existential forgery attack [16], which is even more pronounced under the chosen message attack [2, 17–19]. An incomplete security analysis is one of the key reasons why QS schemes have serious security vulnerabilities. Formal security proofs that reduce the security of QS schemes to the principles of quantum mechanics. The quantum random oracle [20, 21] is used to reduce the security of QS schemes to the quantum no-cloning theorem. The indistinguishability of quantum ciphertext under chosen plaintext attacks [22] provides information-theoretic security for a quantum encryption algorithm. The security models for quantum proxy signatures [23] and arbitration quantum signatures [2] were successively proposed. A formal security proof and a security model for QBS are urgently needed.

In this paper, we demonstrate several security vulnerabilities in the SCF-QBS scheme and verify them using the Qiskit simulator. An improved SCF-QBS (SCF-IQBS) scheme is proposed while retaining the original application scenario. The message space is replaced with a more widely used classical bit space, and the SCF-IQBS scheme provides better quantum bit efficiency. Importantly, we construct the security model of quantum blind signatures based on the security model of classical blind signatures. Through this security model, the security of SCF-IQBS is strictly formalized in games and reduced to the principles of quantum mechanics.

The remainder of this paper is organized as follows: Section 2 reviews the SCF-QBS scheme. Section 3 gives the security analysis of the SCF-QBS scheme. The SCF-IQBS scheme and its security model are proposed in Section 4. Section 5 provides a security analysis. Section 6 gives the conclusion of this paper.

2 | Review of the SCF-QBS Scheme

In this scheme [13], SME is played by Alice and represents an originator signatory. Bob plays CN and represents a verifier. Charlie plays FI and represents a blind signatory. Ideally, Charlie does not have access to Alice's original message. Bob is credible, while Alice and Charlie are not. The χ-state [24] is used to establish a quantum channel, which is represented as

$$\begin{aligned} |\chi\rangle_{3214} &= \frac{1}{2\sqrt{2}}\begin{pmatrix} |0000\rangle - |0011\rangle - |0101\rangle + |0110\rangle + \\ |1001\rangle + |1010\rangle + |1100\rangle + |1111\rangle \end{pmatrix}_{3214} \\ &= \frac{1}{2}\begin{pmatrix} |\Phi^-\rangle|0+\rangle + |\Phi^+\rangle|0-\rangle + \\ |\Psi^-\rangle|1+\rangle + |\Psi^+\rangle|1-\rangle \end{pmatrix}_{3214}. \end{aligned} \tag{1}$$

Where, $|\Phi^\pm\rangle = \frac{1}{\sqrt{2}}(|\phi^+\rangle \pm |\psi^-\rangle)$, $|\Psi^\pm\rangle = \frac{1}{\sqrt{2}}(|\psi^+\rangle \pm |\phi^-\rangle)$, $|\phi^\pm\rangle = \frac{1}{\sqrt{2}}(|00\rangle \pm |11\rangle)$, $|\psi^\pm\rangle = \frac{1}{\sqrt{2}}(|01\rangle \pm |10\rangle)$, and $|\pm\rangle = \frac{1}{\sqrt{2}}(|0\rangle \pm |1\rangle)$. Let $AMB = \{|\Phi^+\rangle, |\Phi^-\rangle, |\Psi^+\rangle, |\Psi^-\rangle\}$ be a set of orthogonal measurement bases. If particle (3, 2) is held by one party, particle (1) is sent to the other party, and particle (4) is sent to a third party, then this scheme can derive the key through the entanglement relation given by Equation (1). In addition, the QOTP algorithm [25] is used to encrypt and decrypt quantum messages. Let M denote the plaintext, S the ciphertext, K the key, E the encryption operation, D the decryption operation, and U a combination of Pauli operators. Then, $M = D_K(S) = \left(U^\dagger\right)^K S = \left(U^\dagger\right)^K E_K(M) = \left(U^\dagger\right)^K U^K M$. U options are X, Z, and Y, where $X|0\rangle(|1\rangle) = |1\rangle(|0\rangle)$, $Z|0\rangle(|1\rangle) = |0\rangle(-|1\rangle)$, $Y = XZ$. The complete QBS scheme consists of four stages: initial, Alice signing, Charlie signing, and verification.

2.1 | Initial Stage

Step 1: Alice and Charlie share the key K_{AC}, and Bob and Charlie share the key K_{BC}, both through a quantum key distribution protocol (QKDP).

Step 2: Bob prepares $\chi^s = \{|\chi(i)\rangle_{3214}\}_{i=1}^q (q > 2n)$ to establish the quantum channel. Then, Bob randomly selects $q - n$ states for channel detection and records their positions. These selected states are labeled u. Bob owns $\{|\chi^u\rangle_{32}\}$, $\{|\chi^u\rangle_1\}$ is sent to Alice by Bob, and $\{|\chi^u\rangle_4\}$ is sent to Charlie by Bob. At the quantum channel detection, Bob announces the positions labeled u. Alice measures all of $\{|\chi^u\rangle_1\}$ with the B_z basis and publishes the results. Alice measures all of $\{|\chi^u\rangle_4\}$ with the B_x basis and publishes the results. Then, Bob measures all of $\{|\chi^u\rangle_{32}\}$ with the AMB basis

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TABLE 1 | The Relationship Between U and K in QOTP Operations.

Choice of K	K Value	Pauli Operator U
$K \in \{K_A\}$	0	Y
	1	X
$K \in \{K_C\}$	0	Y
	1	Z
$K \in \{K_B\}$	00	YY
	01	YZ
	10	XY
	11	XZ
$K \in \{r, K_{AC}, K_{BC}\}$	0	Z
	1	X

and checks the correlation according to Equation (1). If satisfied, the secure channel is successfully established, and the remaining detection particles can be used to perform the remaining steps of the scheme.

2.2 | Alice Signing Stage

Step 3: Let $|P\rangle = \otimes_{i=1}^{n} |p_i\rangle$ denote the two copies of the quantum message of Alice, where $|p_i\rangle = \alpha_i|0\rangle + \beta_i|1\rangle$, $|\alpha_i|^2 + |\beta_i|^2 = 1$.

Step 4: Alice generates a random number r and calculates $|P_x\rangle = E_r(|P\rangle)$, where Table 1 gives the choice of U.

Step 5: Alice encodes the measurement results of the particles in her hand under the B_z basis as K_A. If the result is $|0\rangle(|1\rangle)$, then the corresponding bit of K_A is 0(1).

Step 6: Alice calculates $|R\rangle = E_{K_A}(|P\rangle)$.

Step 7: Alice calculates the signature $|S_A\rangle = E_{K_{AC}}(|P_x\rangle, |R\rangle)$ and sends it to Charlie.

2.3 | Charlie Signing Stage

Step 8: Charlie decrypts $|S_A\rangle$ to get $|P_x\rangle$ and $|R\rangle$ through the key K_{AC}.

Step 9: Charlie encodes the measurement results of the particles in his hand under the B_x basis as K_C. If the result is $|+\rangle(|-\rangle)$, then the corresponding bit of K_C is 0(1).

Step 10: Charlie calculates $|Q\rangle = E_{K_C}(|R\rangle)$.

Step 11: Charlie calculates the signature $|S_C\rangle = E_{K_{BC}}(|P_x\rangle, |Q\rangle)$ and sends it to Bob.

2.4 | Verification Stage

Step 12: Bob announces RE. When Alice gets RE, she announces the random number r.

Step 13: Bob decrypts $|S_C\rangle$ to get $|P_x\rangle$ and $|Q\rangle$ through the key K_{BC}.

Step 14: Bob decrypts $|P_x\rangle$ to get $|P'\rangle$ through the random number r.

Step 15: Bob encodes the measurement results of the particles in his hand under the AMB basis as K_B. If the result is $|\Phi^+\rangle(|\Phi^-\rangle)$, then the corresponding bit of K_B is 01(00). If the result is $|\Psi^+\rangle(|\Psi^-\rangle)$, then the corresponding bit of K_B is 11(10).

Step 16: Bob calculates $|P''\rangle = D_{K_B}(|Q\rangle)$.

Step 17: Bob verifies that $|P'\rangle = |P''\rangle$ holds. If so, the signature is accepted. Otherwise, the verification fails.

3 | Security Analysis of the SCF-QBS Scheme

This section proves that the SCF-QBS scheme is not resistant to the IMR attack and the EM attack. In addition, it is demonstrated through Charlie's view that the privacy-preserving encryption of this scheme is invalid and that he can successfully implement the forgery attack. The attack strategy in question is validated by simulations performed with the IBM Qiskit simulator (Python-3.11.7, Qiskit-1.4.0).

3.1 | The IMR Attack

In the SCF-QBS scheme, the χ-state is used as a quantum resource to establish the channel. To ensure channel security, participants utilize the correlation of χ-states under different measurement bases for detection. However, we found that both external and internal adversaries could escape detection and extract measurement results between participants. Here, we demonstrate that an adversary steals measurement results without being detected through the IMR attack.

In Step 2, Bob sends the first and fourth particle sequences of the χ-state to Alice and Charlie, respectively. In this process, the adversary intercepts the first particle sequence and the fourth particle sequence, respectively. For the first particle sequence, the adversary measures with the B_z basis and records the result, then retransmits the first particle sequence to Alice. For the fourth particle sequence, the adversary measures with the B_x basis and records the results, then retransmits the fourth particle sequence to Charlie. When Bob announces the position of the particles to be measured, Alice measures the particle to be measured with the B_z basis, and Charlie measures the particle to be measured with the B_x basis. The adversary deletes the results of the particle to be measured in the first particle sequence and the fourth particle sequence. Now, the adversary gets the measurement results used to encode K_A and K_C. When Alice and Charlie announce their measurement results, Bob detects correlations through these results with his measurements and finds no errors.

The reason for the success of the adversary is that the first particle is always measured in the B_z basis, whether in the second or the fifth step, and the fourth particle is always measured in the B_x

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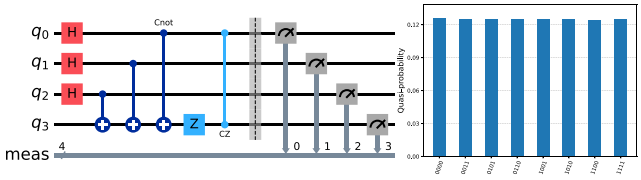

FIGURE 1 | Quantum circuit diagram for the preparation of the χ-state and the results of the measurements.

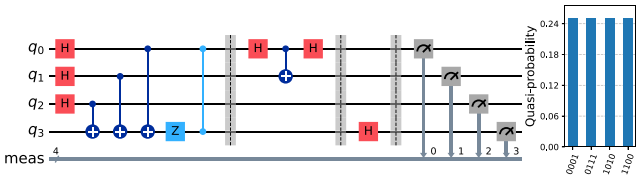

FIGURE 2 | The χ-state is measured in the AMB, B_z and B_x bases. The second part corresponds to the AMB basis, and the third part corresponds to the B_x basis.

basis, whether in the second or the ninth step. The use of the correct measurement basis by the adversary does not introduce errors in the correlation detection of χ-states. Then, this attack is verified in the simulator.

To prepare the χ-state, we have designed a quantum circuit as shown in Figure 1. "H" is a Hadamard gate operation, where $H|0\rangle(|1\rangle) = |+\rangle(|-\rangle)$. "$C_{not}$" is a controlled-not gate operation, where $C_{not}|x, y\rangle = |x, x \oplus y\rangle$, x is the control bit, and $\oplus$ is a bit XOR operation. "CZ" is a controlled-Z gate, where $CZ|1, 1\rangle = |1, -1\rangle$. Each particle of the χ-state is measured using the B_z basis. This quantum circuit diagram is executed enough times to yield the results shown in Figure 1. This result is consistent with Equation (1).

In the SCF-QBS scheme, measurement results are generated in the AMB, B_z, and B_x bases. Therefore, we map the AMB basis and the B_x basis into the B_z basis using gate operations. For the AMB basis, $|\Phi^{\pm}\rangle \xrightarrow{H, C_{not}, H} |01\rangle/|10\rangle$, and $|\Psi^{\pm}\rangle \xrightarrow{H, C_{not}, H} |00\rangle/|11\rangle$. For the B_x basis, $|\pm\rangle \xrightarrow{H} |0\rangle/|1\rangle$. Figure 2 shows the corresponding quantum circuit diagram. The results for the first and fourth particles on the B_z and B_x bases, respectively, are consistent according to Figure 2.

3.2 | The EM Attack

In the EM attack, the adversary entangles the auxiliary particle $\mathscr{E}$ with the target particle or the target quantum system through a U-operation and measures the auxiliary particle to steal information about the target particle or the target quantum system. Here, we demonstrate that an adversary steals a portion of measurement results without being detected through the EM attack.

Let $\mathscr{E} = |0\rangle$, $U = C_{not}$ [26]. The adversary performs the EM attack on the first particle sequence in the channel. Let the first particle be the control bit and let $\mathscr{E}$ be the target bit. Then, after the attack,

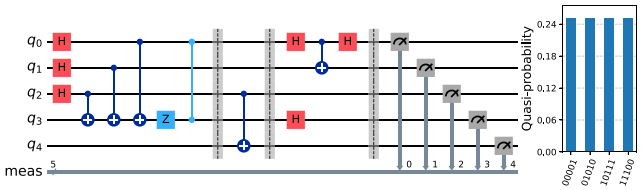

FIGURE 3 | Quantum circuit diagram of the EM attack. The second part is an attack carried out by the adversary.

the quantum system evolves as

$$
\begin{aligned}
&|\chi\rangle_{3214}|0\rangle_{\mathscr{E}} \xrightarrow{C_{not}} \\
&\frac{1}{2\sqrt{2}}\begin{pmatrix} |00000\rangle - |00111\rangle - |01010\rangle + |01101\rangle + \\ |10010\rangle + |10101\rangle + |11000\rangle + |11111\rangle \end{pmatrix}_{3214\mathscr{E}} \\
&= \frac{1}{2}\begin{pmatrix} |\Phi^-\rangle|0+\rangle|0\rangle + |\Phi^+\rangle|0-\rangle|0\rangle + \\ |\Psi^-\rangle|1+\rangle|1\rangle + |\Psi^+\rangle|1-\rangle|1\rangle \end{pmatrix}_{3214\mathscr{E}}.
\end{aligned}
\tag{2}
$$

From Equation (2), we can get that the results of the auxiliary particle $\mathscr{E}$ and the results of the first particle are always consistent in the B_z basis. The correlations of particles (2, 3), particle (1), and particle (4) are consistent with the correlations in Equation (1). This means that the adversary's attack does not introduce errors, and the participants cannot detect the presence of the adversary. However, the fact is that the adversary gets the result of the first particle used to encode K_A through the result of the auxiliary particle.

Figure 3 shows a quantum circuit diagram simulating the EM attack by the adversary. According to the results in Figure 3, the fifth particle $\mathscr{E}$ and the first particle are consistent. There are no errors in the other particles.

3.3 | Privacy-Preserving Encryption is Invalid

The core contribution of the SCF-QBS scheme in SCF scenarios is confidentiality, i.e., protecting the privacy of SMEs. This property is realized through the blinding of the QBS. However, we find that the cryptographic operation used to achieve the blinding is invalid.

Sections 3.1 and 3.2 demonstrate that an adversary can obtain the first-particle measurement results used to encode K_A through the IMR attack and the EM attack. Assuming Charlie is the adversary, he performs an attack to get K_A at Step 2. When Charlie gets $|R\rangle$ at Step 8 by decrypting $|S_A\rangle$, he easily computes $|P\rangle = D_{K_A}(|R\rangle) = D_{K_A}(E_{K_A}(|P\rangle))$. Therefore, privacy-preserving encryption is invalid.

Without loss of generality, let $|P\rangle = |0\rangle|1\rangle|+\rangle|-\rangle$ and $K_A = 0110$. Based on the analysis in Sections 3.1 and 3.2, Charlie can obtain K_A through attacks. Charlie performs the decryption operation shown in Figure 4, and the result is consistent with $|P\rangle$, as shown in Figure 4. $|+\rangle(|-\rangle)$ is mapped to $|0\rangle(|1\rangle)$ by H.

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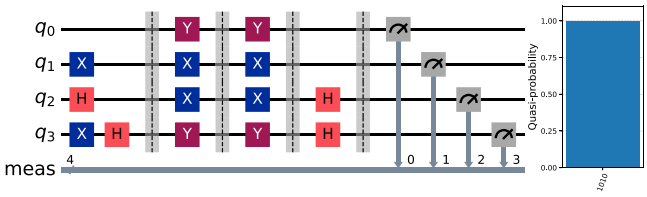

FIGURE 4 | Quantum circuit diagram for decrypting messages. The second part is the encryption operation. The third part is the decryption operation.

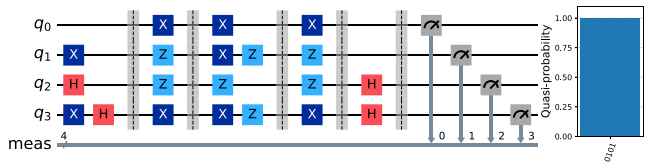

FIGURE 5 | Quantum circuit diagram of the forgery attack. The second part is the encryption operation. The third part is the modification operation. The fourth part is the decryption operation.

3.4 | The Forgery Attack

The forgery attack involves a key-only attack, a known message attack, and an adaptive chosen message attack, and they lead to consequences of universal forgery, selective forgery, and existential forgery [16]. Unforgeability is inherently a fundamental security property of the QBS. However, we found that the SCF-QBS scheme does not fulfill this security property; Charlie can achieve existential forgery.

According to the analysis in [13], Charlie cannot forge a signature due to the existence of the keys K_A and r. Note that r is announced at the verification step and cannot be accessed by Charlie beforehand. But the signature $|S_C\rangle$ passes verification on the condition that $D_r(E_r(|P\rangle)) = |P'\rangle = D_{K_B}(E_{K_C}(E_{K_A}(|P\rangle))) = |P''\rangle$. This means that Charlie only needs to find a valid $D_r(E_r(|\tilde{P}\rangle)) = |\tilde{P}\rangle$ to make the signature $|\tilde{S_C}\rangle$ on $\tilde{P}$ pass verification. According to Sections 3.1 and 3.2, Charlie can obtain K_A and further decrypt $|S_A\rangle$ to obtain $|P_x\rangle$ and $|P\rangle$. Note that whether or not K_A is known does not affect whether Charlie can achieve an existential forgery. This merely demonstrates a powerful combination attack strategy for Charlie, as he can achieve a universal forgery when he knows the state of $|P\rangle$. If Charlie performs the same Pauli operation for each particle of $|P_x\rangle$ and $|P\rangle$, then he realizes $D_r(E_r(|\tilde{P}\rangle)) = D_r(|\tilde{P_x}\rangle) = D_r(\otimes_{i=1}^{L} U_i E_{r^i} |p^i\rangle) = |\tilde{P}\rangle = \otimes_{i=1}^{L} U_i |p^i\rangle$. $|p^i\rangle$ is the ith particle of $|P\rangle$ and r^i is the ith bit of r; L is the length of the quantum message. By the Pauli operation, we have $U_i E_{r^i} = \pm E_{r^i} U_i$, and the minus sign can be ignored due to the quantum state swap test [27] required for verification [16]. Therefore, $D_r(\otimes_{i=1}^{L} E_{r^i} U_i |p^i\rangle) = \otimes_{i=1}^{L} U_i |p^i\rangle$. Charlie sends Bob a forged signature $|\tilde{S_C}\rangle = E_{K_{BC}}(|\tilde{P_x}\rangle, |\tilde{Q}\rangle = E_{K_C}(E_{K_A}(|\tilde{P}\rangle))$, which will pass Bob's verification.

Let $|P\rangle = |0\rangle|1\rangle|+\rangle|-\rangle$ and $r = 1001$. Charlie expects $U = X, XZ, Z, XZ$, and then $U|P\rangle = |1\rangle|0\rangle|-\rangle|+\rangle$. Figure 5 shows the operations on $|P_x\rangle$, and Figure 6 shows the operations on $|P\rangle$. Figures 5 and 6 show the results of $U|P_x\rangle = |\tilde{P_x}\rangle$ and $U|P\rangle = |\tilde{P}\rangle$,

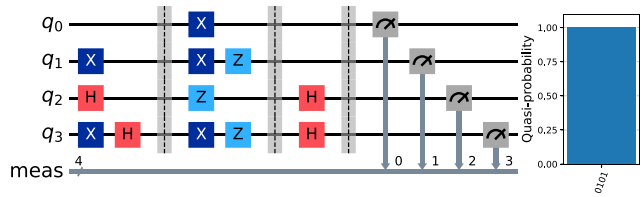

FIGURE 6 | Quantum circuit diagram of the forgery attack. The second part is the modification operation.

respectively. This means that Charlie has successfully found a valid pair of $D_r(|\tilde{P_x}\rangle) = |\tilde{P}\rangle$.

In addition, even if Alice generates a signature, she could deny it because Charlie could forge a signature.

4 | Proposed SCF-IQBS Scheme

In order to maintain the original application scenarios, the original role settings are retained. Let $H(\cdot):\{0,1\}^* \rightarrow \{0,1\}^n$ be the hash function with a uniform output and let $||$ be a string concatenation operation. The bases related to the original definition of the χ-state remain unchanged. Let $Y^{(0)} = H^{(0)} = I$. I is the identity operator.

4.1 | Initial Stage

Step I1: Alice and Charlie share the key K_{AC}, and Bob and Charlie share the key K_{BC}, both through a QKDP.

Step I2: Bob prepares n $|\chi\rangle_{3214}$ states and divides them into four quantum sequences $Q_j = \left(q_j^1, q_j^2, \ldots, q_j^n\right)$. Here, q_j^i is the jth particle of the ith $|\chi\rangle_{3214}$ state, $j = 1, 2, 3, 4$, and $i = 1, 2, 3, \ldots, n$. Then, Bob randomly selects $2n$ decoy particles from the set $\{|0\rangle, |1\rangle, |+\rangle, |-\rangle\}$ and randomly inserts them into Q_1 and Q_4 to obtain Q_1' and Q_4'. Bob sends Q_1' and Q_4' to Alice and Charlie, respectively. When Alice (Charlie) receives Q_1' (Q_4'), Bob announces the position and initial state of the decoy particles in Q_1' (Q_4'). Alice (Charlie) selects the corresponding measurement bases to measure the decoy particles. If the quantum bit error rate exceeds the threshold, the scheme is restarted. Otherwise, Alice (Charlie) deletes the decoy particles to obtain Q_1 (Q_4).

4.2 | Alice Signing Stage

Step S3: Let $M = (m_1, m_2, \ldots, m_n) \in \{0,1\}^n$ denote the message of Alice. First, Alice uses the B_z basis to measure Q_1 to obtain $K_A = (a_1, a_2, \ldots, a_n)$. If q_1^i is $|0\rangle(|1\rangle)$, then a_i is 0(1).

Step S4: Alice calculates $M^1 = (|m_1^1\rangle, |m_2^1\rangle, \ldots, |m_n^1\rangle)$, where $|m_i^1\rangle = H^{(a_i)}|m_i\rangle$. Then, Alice calculates the signature $S_A = \otimes_{i=1}^{n} |sa_i\rangle$, where $|sa_i\rangle = H^{(K_{AC}^{2i})} Y^{(K_{AC}^{2i-1})} |m_i^1\rangle$.

Step S5: Alice calculates $M^2 = H(M||K_A)$. Alice sends $\{S_A, M^2\}$ to Charlie.

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4.3 | Charlie Signing Stage

Step S6: Charlie uses K_{AC} to decrypt $|S_A\rangle$ to get M^1, where $|m_i^1\rangle = (Y^\dagger)^{(K_{AC}^{2i-1})} H^{(K_{AC}^{2i})} |sa_i\rangle$.

Step S7: Charlie uses the B_x basis to measure Q_4 to obtain $K_C = (c_1, c_2, \ldots, c_n)$. If q_4^i is $|+\rangle(|-\rangle)$, then c_i is 0(1).

Step S8: Charlie calculates $M^3 = (|m_1^3\rangle, |m_2^3\rangle, \ldots, |m_n^3\rangle)$, where $|m_i^3\rangle = H^{(c_i)}|m_i^1\rangle$. Then, Charlie calculates the signature $S_C = \otimes_{i=1}^n |sc_i\rangle$, where $|sc_i\rangle = H^{(K_{BC}^{2i})} Y^{(K_{BC}^{2i-1})} |m_i^3\rangle$.

Step S9: Charlie calculates $M^4 = H(M^2||K_C)$. Charlie sends $\{S_C, M^4\}$ to Bob.

4.4 | Verification Stage

Step V10: When Bob receives $\{S_C, M^4\}$, he uses K_{BC} to decrypt $|S_C\rangle$ to get $\widetilde{M^3}$, where $|\widetilde{m_i^3}\rangle = (Y^\dagger)^{(K_{BC}^{2i-1})} H^{(K_{BC}^{2i})} |sc_i\rangle$.

Step V11: Bob uses the AMB basis to measure (Q_3, Q_2) to obtain $K_B = (b_1, b_2, \ldots, b_n)$, $K_A = (a_1, a_2, \ldots, a_n)$, and $K_C = (c_1, c_2, \ldots, c_n)$. Their mappings are as follows:

$$\begin{cases} a_i = 0, b_i = 0, c_i = 0; & \text{if } (q_3^i, q_2^i) = |\Phi^-\rangle \\ a_i = 0, b_i = 1, c_i = 1; & \text{if } (q_3^i, q_2^i) = |\Phi^+\rangle \\ a_i = 1, b_i = 0, c_i = 1; & \text{if } (q_3^i, q_2^i) = |\Psi^-\rangle \\ a_i = 1, b_i = 1, c_i = 0; & \text{if } (q_3^i, q_2^i) = |\Psi^+\rangle \end{cases}. \tag{3}$$

Step V12: Bob calculates $|\widetilde{m_i}\rangle = H^{(b_i)}|\widetilde{m_i^3}\rangle$, measures $|\widetilde{m_i}\rangle$ using the B_z basis to obtain $\widetilde{m_i}$. $\widetilde{M} = (\widetilde{m_1}, \widetilde{m_2}, \ldots, \widetilde{m_n})$.

Step V13: Bob calculates $\widetilde{M^4} = H(H(\widetilde{M}||K_A)||K_C)$. Then, Bob verifies that $\widetilde{M^4} = M^4$. If so, the signature is accepted. Otherwise, the verification fails.

4.5 | A Simple Example

Step EI1: Assume that $K_{AC} = 01100101$ and $K_{BC} = 00101101$.

Step EI2: Bob prepares four $|\chi\rangle_{3214}$ states according to Figure (1) and divides them into four quantum sequences $Q_1 = (q_1^1, q_1^2, q_1^3, q_1^4)$, $Q_2 = (q_2^1, q_2^2, q_2^3, q_2^4)$, $Q_3 = (q_3^1, q_3^2, q_3^3, q_3^4)$, $Q_4 = (q_4^1, q_4^2, q_4^3, q_4^4)$. Here, the eavesdropping detection step is omitted. Alice receives Q_1 sent by Bob. Charlie receives Q_4 sent by Bob.

Step ES3: Let $M = 0101$. First, Alice uses the B_z basis to measure q_1^1, q_1^2, q_1^3, and q_1^4 to obtain $K_A = 1001$ as shown in Figure (7).

Step ES4: Alice calculates M^1, where $|m_i^1\rangle = H^{(a_i)}|m_i\rangle$ corresponds to the first and second parts of Figure (8). Then, Alice calculates the signature S_A, where $|sa_i\rangle = H^{(K_{AC}^{2i})} Y^{(K_{AC}^{2i-1})} |m_i^1\rangle$ corresponds to the third part of Figure (8).

Step ES5: Alice calculates $M^2 = H(0101||1001) = 1001$. Alice sends $\{S_A, M^2\}$ to Charlie.

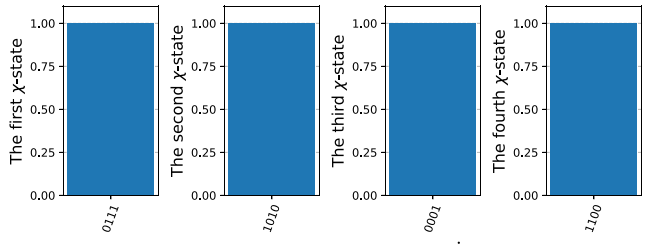

FIGURE 7 | The four $|\chi\rangle_{3214}$ states are the results obtained under the AMB, B_z and B_x bases.

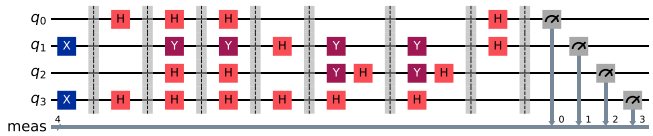

FIGURE 8 | Quantum circuit diagram for the generation and verification of a signature.

Step ES6: Charlie uses K_{AC} to decrypt $|S_A\rangle$ to get M^1, where $|m_i^1\rangle = (Y^\dagger)^{(K_{AC}^{2i-1})} H^{(K_{AC}^{2i})} |sa_i\rangle$ corresponds to the fourth part of Figure (8).

Step ES7: Charlie uses the B_x basis to measure q_4^1, q_4^2, q_4^3, and q_4^4 to obtain $K_C = 0101$ as shown in Figure 7. Here, this measurement operation refers to Figure (2).

Step ES8: Charlie calculates M^3, where $|m_i^3\rangle = H^{(c_i)}|m_i^1\rangle$ corresponds to the fifth part of Figure 8. Then, Charlie calculates the signature S_C, where $|sc_i\rangle = H^{(K_{BC}^{2i})} Y^{(K_{BC}^{2i-1})} |m_i^3\rangle$ corresponds to the sixth part of Figure 8.

Step ES9: Charlie calculates $M^4 = H(1001||0101) = 1000$. Charlie sends $\{S_C, M^4\}$ to Bob.

Step EV10: When Bob receives $\{S_C, M^4\}$, he uses K_{BC} to decrypt $|S_C\rangle$ to get $\widetilde{M^3}$, where $|\widetilde{m_i^3}\rangle = (Y^\dagger)^{(K_{BC}^{2i-1})} H^{(K_{BC}^{2i})} |sc_i\rangle$ corresponds to the seventh part of Figure 8.

Step EV11: Bob uses the AMB basis to measure (q_2^1, q_3^1), (q_2^2, q_3^2), (q_3^1, q_3^3), and (q_2^4, q_3^4) to obtain $K_B = 1100$ as shown in Figure 7. Here, this measurement operation refers to Figure 2. Bob obtains $K_A = 1001$ and $K_C = 0101$ according to K_B and Equation (3).

Step EV12: Bob calculates $|\widetilde{m_i}\rangle = H^{(b_i)}|\widetilde{m_i^3}\rangle$, measures $|\widetilde{m_i}\rangle$ using the B_z basis to obtain $\widetilde{m_i}$. These operations correspond to the eighth part of Figure 8. $\widetilde{M} = 0101$ as shown in Figure 9.

Step EV13: Bob calculates $\widetilde{M^4} = H(H(0101||1001)||0101) = 1000$. Therefore, this signature passes verification because $\widetilde{M^4} = M^4$.

4.6 | Security Model

An effective QBS scheme needs to satisfy correctness, blindness, unforgeability, and undeniability. Based on the security model of classical blind signatures [28, 29], we present a security model

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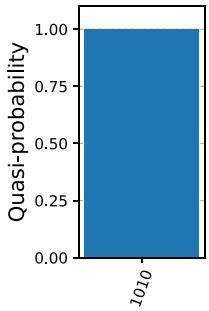

FIGURE 9 | The results for $\widetilde{M}$.

for blindness that is applicable to QBS. In order to achieve high standards of security for QBS, existential unforgeability under the adaptive chosen message attack was selected. In addition, some tools used for proofs are defined and described.

4.6.1 | Correctness

For a QBS of any message, if all participants honestly execute the steps of the scheme, then this QBS must be successfully verified.

4.6.2 | Blindness

The QBS requires that any signed message cannot be accessed by the blind signer. The blindness of SCF-IQBS is formally defined by a game between a malicious adversary $\mathcal{A}$ and an honest challenger $\mathcal{C}$.

- $\mathcal{C}$ transmits the public parameters in the scheme to $\mathcal{A}$ and initiates the initial steps.
- $\mathcal{A}$ selects two independent messages M_0 and M_1 and submits them to $\mathcal{C}$.
- $\mathcal{C}$ obtains a $\upsilon \in \{0,1\}$ by tossing a coin. $\mathcal{C}$ generates $\{S_A, M^2\}$ about M_υ and returns them to $\mathcal{A}$.
- $\mathcal{A}$ guesses a $\upsilon' \in \{0,1\}$. If $\upsilon' = \upsilon$, then $\mathcal{A}$ wins with an advantage.

$$Adv(\mathcal{A}) = \left|\Pr[\upsilon' = \upsilon] - \frac{1}{2}\right|. \tag{4}$$

4.6.3 | Unforgeability

The SCF-IQBS scheme has two types of polynomial-time adversaries for forged signatures. $\mathcal{A}_4$ is employed to simulate a malicious Alice attempting to forge blind signatures. It is worth noting that if neither $\mathcal{A}_4$ can forge a signature, then an external adversary who does not know any secret parameters cannot forge a signature either. $\mathcal{A}_4$ can launch attacks such as intercept and replay, EM, and tampering on the quantum channel. $\mathcal{A}_4$ can launch attacks such as intercept, tampering, and replay on the classical channel. $\mathcal{C}$ acts as Charlie and Bob in response to queries from $\mathcal{A}_4$, and they interact through the following queries.

- Initialization phase. The keys are generated through the QKDP and $\mathcal{A}_4$ obtains the public parameters.
- **Quantumsequencequery**(M_t, t). $\mathcal{C}$ maintains a list $List - Q$ containing information $(M_t, t, \Gamma, K_{At}, K_{Bt}, K_{Ct}, PosQ_{4t})$. M_t is the message chosen by $\mathcal{A}_4$, t is used to mark a query. When the query is triggered, $\mathcal{C}$ checks whether (M_t, t) exists in $List - Q$. If (M_t, t) does not exist, $\mathcal{C}$ generates n $|\chi\rangle_{3214}$ states and n decoy particles. The generation of Q_j is as in Step I2. $\mathcal{C}$ randomly inserts these decoy particles into Q_{4t} to obtain Q'_{4t}, and records the position and state of the decoy particles as $PosQ_{4t}$. $\mathcal{C}$ returns Q'_{4t} and Q_{1t} to $\mathcal{A}_4$. $\mathcal{C}$ measures particles (3,2) using the AMB basis to obtain K_{Ct} and infer K_{Bt} and K_{At} and $\Gamma = 1$. If (M_t, t) is present, $\mathcal{C}$ checks whether Γ equals 1. If $\Gamma = 1$, $\mathcal{C}$ returns $NULL$ to $\mathcal{A}_4$. Otherwise, $\mathcal{C}$ prepares Q_{4t} and Q_{1t} based on the recorded K_{Bt} and K_{At}. $\mathcal{C}$ prepares n decoy particles and randomly inserts into Q_{4t} to obtain Q'_{4t}, recording the position and state of the decoy particles as $PosQ_{4t}$. $\mathcal{C}$ returns Q'_{4t} and Q_{1t} to $\mathcal{A}_4$, and sets $\Gamma = 1$. When $\mathcal{A}_4$ executes a query and obtains a particle sequence, $\mathcal{A}_4$ must return a quantum sequence to $\mathcal{C}$, identical in form to Q'_{4t}, for the eavesdropping detection. Note that the particles (2, 3) are measured first, and the quantum sequences Q_{4t} and Q_{1t} are prepared according to K_{Bt} and K_{At}, which does not produce a difference in the view of $\mathcal{A}_4$. Because the entanglement correlation of $|\chi\rangle_{3214}$ originates from the particle subsystems (2, 3) and (1, 4), the reduced density matrix for particles (1, 4) is $I/4$. $NULL$ indicates an invalid query, ensuring that $\mathcal{A}_4$ may query each quantum sequence only once.
- **Hashquery1**(M_t, K_{At}). $\mathcal{C}$ maintains a list $List - H1$ containing (M_t, K_{At}, M_t^2). $\mathcal{A}_4$ submits M_t and K_{At}. If M_t and K_{At} exist in $List - H1$, $\mathcal{C}$ returns the corresponding M_t^2 to $\mathcal{A}_4$. If M_t and K_{At} do not exist in $List - H1$, $\mathcal{C}$ selects a random number as M_t^2 and returns it to $\mathcal{A}_4$ and records the pertinent parameters.
- **Hashquery2**(M_t^2, K_{Bt}). $\mathcal{C}$ maintains a list $List - H2$ containing (M_t^2, K_{Bt}, M_t^4). $\mathcal{A}_4$ submits M_t^2 and K_{Bt}. If M_t^2 and K_{Bt} exist in $List - H2$, $\mathcal{C}$ returns the corresponding M_t^4 to $\mathcal{A}_4$. If M_t^2 and K_{Bt} do not exist in $List - H2$, $\mathcal{C}$ selects a random number as M_t^4 and returns it to $\mathcal{A}_4$ and records the pertinent parameters.
- **Signaturequery**(M_t, t). $\mathcal{C}$ maintains a list $List - S$ containing $(M_t, t, K_{At}, K_{Bt}, M_t^2, M_t^4)$. $\mathcal{C}$ receives M_t sent from $\mathcal{A}_4$. $\mathcal{C}$ checks whether (M_t, t) exists in $List - S$. If it exists, $\mathcal{C}$ returns M_t^2 and M_t^4 to $\mathcal{A}_4$. Otherwise, $\mathcal{C}$ checks whether (M_t, t) exists in $List - Q$. If (M_t, t) does not exist, $\mathcal{C}$ adds the relevant record for (M_t, t) to $List - Q$ and sets $\Gamma = 0$. If (M_t, t) exists, $\mathcal{C}$ calculates S_{Ct} using K_{At}, K_{Bt}, and M_t. Then $\mathcal{C}$ checks whether M_t exists in $List - H1$. If M_t does not exist, $\mathcal{C}$ adds the relevant record for M_t to $List - H1$. Otherwise, $\mathcal{C}$ selects M_t^2. Finally, $\mathcal{C}$ checks whether M_t^2 exists in $List - H2$. If M_t^2 does not exist, $\mathcal{C}$ adds the relevant record for M_t^2 to $List - H2$. Otherwise, $\mathcal{C}$ returns S_{Ct}, M_t^2, and M_t^4 to $\mathcal{A}_4$.

Forge signature. $\mathcal{A}_4$ tries to forge a signature following a series of queries. $\mathcal{A}_4$ can perform q_Q **Quantumsequencequery**, q_{H1} **Hashquery1**, q_{H2} **Hashquery2** and q_S **Signaturequery**, where $q_H = q_{H1} + q_{H2}$. $\mathcal{A}_4$ outputs $(\widetilde{M}, \widetilde{S_C}, \widetilde{M^4})$. If $\widetilde{M}$ was not submitted to any signature queries and this signature passes verification under the parameters of some query recorded in $List - Q$, then $\mathcal{A}_4$ wins.

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4.6.4 | Indistinguishability

An information-theoretically secure quantum encryption algorithm should satisfy the indistinguishability of quantum ciphertexts [22].

Definition 1. Let $E_*(\cdot)$ be a quantum encryption algorithm with G as its internal coin-tossing operation. The algorithm achieves indistinguishability if, for every positive polynomial $p(\cdot)$, any family of quantum distinguisher circuits $\{C_n\}_{n\in\mathbb{N}}$, all sufficiently large $n \in \mathbb{N}$, and any two inputs $x, y \in \{0,1\}^{p(n)}$, the following holds:

$$\left|\Pr\left[C_n\left(E_{G(1^n)}(x)\right) = 1\right] - \Pr\left[C_n\left(E_{G(1^n)}(y)\right) = 1\right]\right| < 1/p(n). \quad (5)$$

Yang et al. [22] proved that the indistinguishability is captured by the trace distance between quantum ciphertexts.

Theorem 1. *For quantum ciphertexts $E_*(x)$, $E_*(y)$, and $p(\cdot)$. If the condition in Definition 1 holds, then the trace distance Dist between their quantum ciphertext density operators ρ_* satisfies:*

$$Dist(\rho_{E_*(x)}, \rho_{E_*(y)}) < 1/p(n). \quad (6)$$

4.6.5 | Random oracle model

Definition 2. A random oracle is an idealizzed model possessing the following properties [30]:

Randomness: For each different input, the output is uniformly distributed randomly over $\{0,1\}^n$.

Consistency: The same input always produces the same output.

Validity: The output of any input is computable in polynomial time.

In practical protocol security analysis, hash functions are idealized as this random oracle model that can be accessed by any participant within polynomial time.

5 | Security Analysis

In QS schemes, some entangled quantum states are distributed using a decoy particle detection mechanism. An authenticated classical channel is used to publish information about the decoy particles. The security of QS is built on the assumption that the classical channel is authenticated. In addition, the QKDP is also considered to be unconditionally secure. Based on this, we perform a formal security proof for the SCF-IQBS scheme.

5.1 | Analysis of the Correctness

Theorem 2. *The SCF-IQBS scheme satisfies the correctness.*

Proof.

$$\begin{aligned}
|m_i\rangle &= H^{(b_i)}(Y^\dagger)^{(K_{BC}^{2i-1})}H^{(K_{BC}^{2i})}|sc_i\rangle \\
&= H^{(b_i)}(Y^\dagger)^{(K_{BC}^{2i-1})}H^{(K_{BC}^{2i}\oplus K_{BC}^{2i})}Y^{(K_{BC}^{2i-1})}|m_i^3\rangle \\
&= H^{(b_i)}H^{(c_i)}(Y^\dagger)^{(K_{AC}^{2i-1})}H^{(K_{AC}^{2i}\oplus K_{AC}^{2i})}Y^{(K_{AC}^{2i-1})}H^{(a_i)}|m_i\rangle \\
&= H^{(b_i)}H^{(c_i)}H^{(a_i)}|m_i\rangle \\
&= H^{(b_i\oplus c_i\oplus a_i)}|m_i\rangle \\
&= |m_i\rangle.
\end{aligned} \quad (7)$$

Therefore, a message obtained via quantum signature yields the same result as M^4 when computed through $H(H(M||K_A)||K_C)$. □

5.2 | Analysis of the Quantum Sequence Distribution

5.2.1 | The Indistinguishability of Q_1' and Q_4'

Theorem 3. *Q_1 and Q_4 both satisfy the indistinguishability property.*

Proof. Q_u' is generated by randomly mixing Q_u and randomly prepared decoy particles. Based on the density operator of the χ-state given by Equation (8), we take partial traces for particles (324) and particles (321), respectively.

$$\begin{aligned}
\rho_\chi &= |\chi\rangle\langle\chi|_{3214} \\
&= \frac{1}{4}\begin{pmatrix} |\Phi^-\rangle\langle\Phi^-||0+\rangle\langle 0+| + |\Phi^+\rangle\langle\Phi^+||0-\rangle\langle 0-| + \\ |\Psi^-\rangle\langle\Psi^-||1+\rangle\langle 1+| + |\Psi^+\rangle\langle\Psi^+||1-\rangle\langle 1-| \end{pmatrix}_{3214}.
\end{aligned} \quad (8)$$

$$\rho_1 = \mathrm{Tr}_1(\rho_\chi) = \frac{1}{2}(|0\rangle\langle 0| + |1\rangle\langle 1|) = \frac{I}{2}. \quad (9)$$

$$\rho_4 = \mathrm{Tr}_4(\rho_\chi) = \frac{1}{2}(|+\rangle\langle +| + |-\rangle\langle -|) = \frac{I}{2}. \quad (10)$$

$$\rho_d = \frac{1}{4}(|0\rangle\langle 0| + |1\rangle\langle 1| + |+\rangle\langle +| + |-\rangle\langle -|) = \frac{I}{2}. \quad (11)$$

Equation (9) gives the reduced density operator for the first particle. Equation (10) gives the reduced density operator for the fourth particle. Equation (11) gives the density operator of the decoy particles. Therefore, the density operators for Q_1' and Q_4' are given by Equation (12) and Equation (13), respectively.

$$\rho_{Q_1'} = \left(\frac{n}{n+n}\rho_1 + \frac{n}{n+n}\rho_d\right)^{\otimes(2n)} = \left(\frac{I}{2}\right)^{\otimes 2n}. \quad (12)$$

$$\rho_{Q_4'} = \left(\frac{n}{n+n}\rho_4 + \frac{n}{n+n}\rho_d\right)^{\otimes(2n)} = \left(\frac{I}{2}\right)^{\otimes 2n}. \quad (13)$$

Let $\widetilde{Q_u'}$ be another quantum sequence. Clearly, $\rho_{\widetilde{Q_u'}} = (\frac{I}{2})^{\otimes 2n}$.According to Equation (6), we have $Dist(\rho_{Q_u'}, \rho_{\widetilde{Q_u'}}) = 0$. □

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5.2.2 | Analysis of the Intercept-Resend Attack

In the intercept-resend attack, $\mathcal{A}$ has two attack modes. Due to the indistinguishability of Q'_u and the existence of decoy particles, $\mathcal{A}$ needs to guess the initial state of the decoy particles in order to evade detection. It is worth noting that the indistinguishability of Q'_u ensures that the advantage gained by $\mathcal{A}$ does not exceed a random guess. If $\mathcal{A}$ intercepts Q'_u and replaces the particles within it. The probability of success of this attack is $(\frac{2}{4} \times \frac{1}{2} + \frac{1}{4})^n$. If $\mathcal{A}$ intercepts Q'_u and measures it. The probability of success of this attack is $(\frac{1}{2} \times 1 + (1 - \frac{1}{2}) \times \frac{1}{2})^n$.

5.2.3 | Analysis of the EM Attack

According to Section 3.2, for the decoy particles, described after the $\mathcal{A}$ attack as

$$\begin{cases} U|0\rangle|\mathscr{E}\rangle = a|0\rangle|\mathscr{E}_{00}\rangle + a^*|1\rangle|\mathscr{E}_{01}\rangle \\ U|1\rangle|\mathscr{E}\rangle = b^*|0\rangle|\mathscr{E}_{10}\rangle + b|1\rangle|\mathscr{E}_{11}\rangle \\ U|+\rangle|\mathscr{E}\rangle = \frac{1}{2}\begin{pmatrix} a|0\rangle|\mathscr{E}_{00}\rangle + a^*|1\rangle|\mathscr{E}_{01}\rangle + \\ b^*|0\rangle|\mathscr{E}_{10}\rangle + b|1\rangle|\mathscr{E}_{11}\rangle \end{pmatrix} \\ U|-\rangle|\mathscr{E}\rangle = \frac{1}{2}\begin{pmatrix} a|0\rangle|\mathscr{E}_{00}\rangle + a^*|1\rangle|\mathscr{E}_{01}\rangle - \\ b^*|0\rangle|\mathscr{E}_{10}\rangle - b|1\rangle|\mathscr{E}_{11}\rangle \end{pmatrix} \end{cases}. \tag{14}$$

where $|a|^2 + |a^*|^2 = 1, |b^*|^2 + |b|^2 = 1$. When $i = j$, $\langle\mathscr{E}_i|\mathscr{E}_j\rangle = 1$, $i, j \in \{00, 01, 10, 11\}$. When $i \neq j$, $\langle\mathscr{E}_i|\mathscr{E}_j\rangle = 0$. If the initial state of the decoy particles is not changed, then $a^* = b^* = 0$ should be made. Equation (14) becomes

$$\begin{cases} U|0\rangle|\mathscr{E}\rangle = a|0\rangle|\mathscr{E}_{00}\rangle \\ U|1\rangle|\mathscr{E}\rangle = b|1\rangle|\mathscr{E}_{11}\rangle \\ U|+\rangle|\mathscr{E}\rangle = \frac{1}{2}\{|+\rangle(a|\mathscr{E}_{00}\rangle + b|\mathscr{E}_{11}\rangle)\} + \\ \qquad \frac{1}{2}\{|-\rangle(a|\mathscr{E}_{00}\rangle - b|\mathscr{E}_{11}\rangle)\} \\ U|-\rangle|\mathscr{E}\rangle = \frac{1}{2}\{|+\rangle(a|\mathscr{E}_{00}\rangle - b|\mathscr{E}_{11}\rangle)\} + \\ \qquad \frac{1}{2}\{|-\rangle(a|\mathscr{E}_{00}\rangle + b|\mathscr{E}_{11}\rangle)\} \end{cases}. \tag{15}$$

The attack of $\mathcal{A}$ on $\{|+\rangle, |-\rangle\}$ changes its initial state with a probability of $\frac{1}{2}$ obtained from Equation (15). If $a|\mathscr{E}_{00}\rangle = b|\mathscr{E}_{11}\rangle$, the attack does not change the initial state of the decoy particle. For the χ-state, the quantum system is described as

$$\begin{aligned} & U|\chi\rangle_{3214}|\mathscr{E}\rangle_{e1}|\mathscr{E}\rangle_{e2} \\ &= \frac{1}{2}\begin{bmatrix} |\Phi^-\rangle_{32}(U|0\rangle_1|\mathscr{E}\rangle_{e1})(U|+\rangle_4|\mathscr{E}\rangle_{e2}) + \\ |\Phi^+\rangle_{32}(U|0\rangle_1|\mathscr{E}\rangle_{e1})(U|-\rangle_4|\mathscr{E}\rangle_{e2}) + \\ |\Psi^-\rangle_{32}(U|1\rangle_1|\mathscr{E}\rangle_{e1})(U|+\rangle_4|\mathscr{E}\rangle_{e2}) + \\ |\Psi^+\rangle_{32}(U|1\rangle_1|\mathscr{E}\rangle_{e1})(U|-\rangle_4|\mathscr{E}\rangle_{e2}) \end{bmatrix} \\ &= \frac{1}{2}\begin{bmatrix} a^2(|\Phi^-\rangle|0+\rangle)_{3214}(|\mathscr{E}_{00}\rangle|\mathscr{E}_{00}\rangle)_{e1e2} + \\ a^2(|\Phi^+\rangle|0-\rangle)_{3214}(|\mathscr{E}_{00}\rangle|\mathscr{E}_{00}\rangle)_{e1e2} + \\ a^2(|\Psi^-\rangle|1+\rangle)_{3214}(|\mathscr{E}_{00}\rangle|\mathscr{E}_{00}\rangle)_{e1e2} + \\ a^2(|\Psi^+\rangle|1-\rangle)_{3214}(|\mathscr{E}_{00}\rangle|\mathscr{E}_{00}\rangle)_{e1e2} \end{bmatrix}. \end{aligned} \tag{16}$$

From Equation (16), it can be concluded that the state of the auxiliary particle is fixed and $\mathcal{A}$ cannot extract any information from the auxiliary particle. Therefore, the probability that $\mathcal{A}$ successfully obtains valid information without introducing errors is $(\frac{1}{2} \times 1 + (1 - \frac{1}{2}) \times \frac{1}{2})^n$.

5.3 | Analysis of the Blindness

Theorem 4. *M^1 satisfies the indistinguishability property.*

Proof. According to Step S4, $|m_i^1\rangle = H^{(a_i)}|m_i\rangle$. Therefore, $|m_i\rangle \in \{|0\rangle, |1\rangle\}$ and we have

$$\begin{aligned} \rho_{M^1} &= \frac{1}{2^n}\bigotimes_{i=1}^{n}\sum_{a_i\in\{0,1\}}\left(H^{(a_i)}|m_i\rangle\langle m_i|H^{(a_i)}\right) \\ &= \frac{1}{2^n}\bigotimes_{i=1}^{n}\sum\begin{pmatrix} |0\rangle\langle 0| & H|1\rangle\langle 1|H \\ H|0\rangle\langle 0|H & |1\rangle\langle 1| \end{pmatrix} \\ &= \left(\frac{I}{2}\right)^{\otimes n}. \end{aligned} \tag{17}$$

Let $\widetilde{M_1}$ be the quantum ciphertext of another $\widetilde{M}$. Clearly, $\rho_{\widetilde{M_1}} = (\frac{I}{2})^{\otimes n}$. According to Equation (6), we have $Dist(\rho_{M_1}, \rho_{\widetilde{M_1}}) = 0$. □

Theorem 5. *The SCF-IQBS satisfies blindness under the random oracle model.*

Proof. $\mathcal{A}_1$ acts as a malicious blind signer in this section. C obtains a $v \in \{0, 1\}$ by tossing a coin and generates $\{S_A, M^2\}$ about M_v. $\mathcal{A}_1$ holds the key K_{AC}. $M_v^1 = H^{(K_A)}|M_v\rangle$, $M_v^2 = H(M_v||K_A)$. $\mathcal{A}_1$ calculates M_v through M_v^2, which is equivalent to solving the hash function pre-mapping problem, with the probability of $\mathcal{A}_1$ success being negligible. M_v^1 satisfying the indistinguishability implies that the density operator of any quantum cryptographic ciphertext is $(I/2)^{\otimes n}$. If the advantage gained by $\mathcal{A}_1$ in extracting plaintext M_v from M_v^1 is non-negligible, then $\mathcal{A}_1$ can be exploited by an algorithm to accurately distinguish unknown non-orthogonal quantum state sequences with a non-negligible advantage. $\mathcal{A}_1$ steals K_A, which is equivalent to compromising the security of the quantum sequence distribution. Section 5.2 has demonstrated the security of quantum sequence distribution; if the advantage of $\mathcal{A}_1$ is non-negligible, then $\mathcal{A}_1$ can be exploited by an algorithm to accurately distinguish unknown non-orthogonal quantum state sequences with a non-negligible advantage. Thus, the probability that $\mathcal{A}_1$ guesses v should not exceed 1/2, that is, $\Pr[v' = v] = 1/2 + \mathrm{negl}(n)$. $\mathrm{negl}(n)$ is a negligible function. The advantage that $\mathcal{A}_1$ has in winning this game is $Adv(\mathcal{A}_1) = \left|\Pr[v' = v] - \frac{1}{2}\right| = \mathrm{negl}(n)$. □

5.4 | Analysis of the Unforgeability

Theorem 6. *The SCF-IQBS scheme satisfies the existential unforgeability under the adaptive chosen message attack in the random oracle.*

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$\mathcal{C}$ needs to accept a challenge before the game begins. $\mathcal{C}$ receives $S^* \in \{|0\rangle, |1\rangle, |+\rangle, |-\rangle\}^n$ from outside. $\mathcal{C}$ needs to output the state of each bit of S^*.

Lemma 1. *For any polynomial-time adversary $\mathcal{A}_4$ that forges a signature in the SCF-IQBS scheme with a non-negligible advantage ϵ, there exists $\mathcal{C}$ that can use $\mathcal{A}_4$ as a subroutine to solve the accepted challenge with a non-negligible advantage* $(\frac{q_H}{2^n} + \frac{{q_H}^2}{2^{n+1}} + \frac{1}{q_S} \cdot Adv_{S^*}^{Distinguish})$.

Proof. Based on the above lemma, we prove the unforgeability of the SCF-IQBS by defining four games.

Game $\mathcal{G}_0$: This game simulates attacks by $\mathcal{A}_4$ on the SCF-IQBS scheme in the real world. The interaction between $\mathcal{A}_4$ and $\mathcal{C}$ is based on the real SCF-IQBS scheme. The advantage of $\mathcal{A}_4$ is

$$Adv_{\mathcal{G}_0}^{SCF-IQBS} = \epsilon. \tag{18}$$

Game $\mathcal{G}_1$: Same as **Game** $\mathcal{G}_0$, the only difference is that the hash function in the SCF-IQBS scheme is replaced with a random oracle. The birthday paradox [31] provides an upper bound on the probability that an adversary finds a hash collision or correctly predicts an output after q_H queries. The difference in advantage between $\mathcal{A}_4$ in **Game** $\mathcal{G}_1$ and **Game** $\mathcal{G}_0$ satisfies

$$\left|Adv_{\mathcal{G}_0} - Adv_{\mathcal{G}_1}\right| \leq \frac{q_H}{2^n} + \frac{{q_H}^2}{2^{n+1}}. \tag{19}$$

Game $\mathcal{G}_2$: Same as **Game** $\mathcal{G}_1$, the only difference is that $\mathcal{C}$ replacing Q_4 in **Quantumsequencequery** with a quantum sequence of length n, randomly selected from $\{|+\rangle, |-\rangle\}$. The sole distinction between **Game** $\mathcal{G}_1$ and **Game** $\mathcal{G}_2$ lies in Q_4. Q_4' in $\mathcal{G}_1$ and Q_4' in $\mathcal{G}_2$ are identical from the view of $\mathcal{A}_4$, since both $\rho_{Q_4'}$ in $\mathcal{G}_1$ and $\rho_{Q_4'}$ in $\mathcal{G}_2$ are $(I/2)^{\otimes 2n}$. The difference in advantage between $\mathcal{A}_4$ in **Game** $\mathcal{G}_2$ and **Game** $\mathcal{G}_1$ satisfies

$$Adv_{\mathcal{G}_1} = Adv_{\mathcal{G}_2}. \tag{20}$$

Game $\mathcal{G}_3$: Same as **Game** $\mathcal{G}_2$, the only difference is that $\mathcal{C}$ randomly selects a k and directly employs S^* as S_{Ck} to return to $\mathcal{A}_4$ in **Signaturequery**. $1 \leq k \leq q_S$. If $\mathcal{A}_4$ wins the game using the kth S_{Ck}, there exist quantum operations U_i such that $|\widetilde{M}\rangle = \bigotimes_{i=1}^n |\widetilde{m_i}\rangle = \bigotimes_{i=1}^n H^{(bk_i)}(Y^\dagger)^{(K_{BC}^{2i-1})}H^{(K_{BC}^{2i})}|\widetilde{sc_i}\rangle = \bigotimes_{i=1}^n U_i|s_i^*\rangle$. $\mathcal{A}_4$ has obtained the evolution process of $\widetilde{S_C}$ and the forgery operation U_i. This means that $\mathcal{A}_4$ is familiar with the state of S^*. Let $Adv_{S^*}^{Distinguish}$ denote the advantage of accurately distinguishing S^*. The difference in advantage between $\mathcal{A}_4$ in **Game** $\mathcal{G}_3$ and **Game** $\mathcal{G}_2$ satisfies

$$\left|Adv_{\mathcal{G}_2} - Adv_{\mathcal{G}_3}\right| \leq \frac{1}{q_S} \cdot Adv_{S^*}^{Distinguish}. \tag{21}$$

Through the hybrid argument and $\mathcal{G}_0 \sim \mathcal{G}_3$, we have Equation (22).

$$\epsilon \leq \frac{q_H}{2^n} + \frac{{q_H}^2}{2^{n+1}} + \frac{1}{q_S} \cdot Adv_{S^*}^{Distinguish}. \tag{22}$$

When n is sufficiently large. If there exists $\mathcal{A}_4$ that successfully forges a signature with a non-negligible advantage ϵ, then there exists a polynomial time $\mathcal{C}$ that uses $\mathcal{A}_4$ as a subprogram to solve the accepted challenge with a non-negligible advantage $\frac{q_H}{2^n} + \frac{{q_H}^2}{2^{n+1}} + \frac{1}{q_S} \cdot Adv_{S^*}^{Distinguish}$. However, an unknown non-orthogonal quantum state sequence cannot be accurately distinguished. □

5.5 | Analysis of the undeniability

In Section 5.4, we prove that the SCF-IQBS scheme is resistant to the existential forgery under the adaptive chosen message attack. This means that only legitimate Alice and Charlie can generate a valid signature through legitimate operations. Therefore, when a signature passes verification, it can only have been generated by Alice and Charlie. Alice or Charlie cannot deny the fact that they signed.

6 | Discussion and Conclusion

In the SCF-QBS scheme, both external and internal adversaries can launch the IMR attack and the EM attack, and the participants cannot effectively detect these attacks. This also means that adversaries can capture measurement results of the χ-states used to encode the key, thus causing the operations used to encrypt the original message to be ineffective against the internal adversary. In addition, the blind signer can achieve existential forgery upon receiving a valid signature from the signature originator. In terms of quantum resource consumption, the SCF-QBS scheme consumes $2n$ χ-states in the initial stage and $2n$ single-particle states in the signing stage. Therefore, signing an n-bit message consumes a total of $10n$ quantum bits. In addition, the quantum state swap test is required during the verification stage to verify the legitimacy of a signature, and an accurate verification result may consume more quantum bits [32].

In the SCF-IQBS scheme, all security properties and adversary capabilities are defined by the security model, and the security of the scheme is captured by formal security analysis. In particular, blindness is reduced to distinguishing between two quantum ciphertexts derived from different plaintexts and solving the preimage of a hash function; unforgeability is reduced to distinguishing an unknown non-orthogonal quantum sequence. In terms of quantum resource consumption, the SCF-IQBS scheme consumes n χ-states and $2n$ decoy particles in the initial stage and n single-particle states in the signing stage. Therefore, signing an n-bit message consumes a total of $7n$ quantum bits. Furthermore, the signed message space is the classic bit space that is more widely applicable, and the quantum state swap test is not required during the verification phase.

In summary, we have demonstrated several security vulnerabilities in the SCF-QBS scheme and verified them using the Qiskit simulator. Subsequently, the SCF-IQBS scheme was proposed, and the corresponding security model established. Compared to the original scheme, the SCF-IQBS scheme consumes fewer quantum bits and provides strict security. The enhanced security and efficiency features of the SCF-IQBS scheme render it suitable for scenarios requiring signatures within quantum-secure environments whilst safeguarding privacy. Potential applications may include, but are not limited to, electronic voting systems, privacy-preserving digital rights management, and sealed-bid

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auction scenarios. These scenarios similarly require the two core properties of blindness and unforgeability.

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Conflicts of Interest

The authors declare no conflict of interest.

Data Availability Statement

Data sharing is not applicable to this article as no new data were created or analyzed in this study

References

1. P. W. Shor, "Polynomial-Time Algorithms for Prime Factorization and Discrete Logarithms on a Quantum Computer," *SIAM Review* 41, no. 2 (1999): 303–332, https://doi.org/10.1137/S0036144598347011.

2. T. Zhang, X. Xin, L. Sun, C. Li, and F. Li, "Secure Quantum Signature Scheme Without Entangled State," *Quantum Information Processing* 23, no. 2 (2024): 49, https://doi.org/10.1007/s11128-024-04257-5.

3. R. Han, K.-J. Zhang, K.-C. Hou, L. Zhang, and X. Zhao, "A New Quantum Multi-Party Signature Protocol Based on Snop States Without Arbitrator," *Physica A: Statistical Mechanics and its Applications* 611 (2023): 128453, https://doi.org/10.1016/j.physa.2023.128453.

4. X. Wei, S. Luo, D. Meng, D. Lu, and Z. Li, "A Novel Quantum Multiparty Blind Signature Scheme Based on χ State," *International Journal of Quantum Information* 22, no. 07 (2024): 2450011, https://doi.org/10.1142/S0219749924500114.

5. J. Cao, X. Xin, C. Li, and F. Li, "Security Analysis and Improvement of a Blind Semi-Quantum Signature," *International Journal of Theoretical Physics* 62, no. 4 (2023): 87, https://doi.org/10.1007/s10773-023-05350-8.

6. X. Wen, X. Niu, L. Ji, and Y. Tian, "A Weak Blind Signature Scheme Based on Quantum Cryptography," *Optics Communications* 282, no. 4 (2009): 666–669, https://doi.org/10.1016/j.optcom.2008.10.025.

7. H. Zhu, Y. Zhang, and Z. Li, "Efficient Quantum Blind Signature Scheme Based on Quantum Fourier Transform," *International Journal of Theoretical Physics* 60 (2021): 2311–2321, https://doi.org/10.1007/s10773-021-04854-5.

8. C. Xia, H. Li, and J. Hu, "A Semi-Quantum Blind Signature Protocol Based on Five-Particle Ghz State," *European Physical Journal Plus* 136, no. 6 (2021): 633, https://doi.org/10.1140/epjp/s13360-021-01605-7.

9. Q. Yuwei, C. Sihao, L. Na, and B. Qian, "Quantum Voting Protocol Based on Blind Signature," in *The International Conference on Natural Computation, Fuzzy Systems and Knowledge Discovery* (Springer, 2022), 551–558, https://doi.org/10.1007/978-3-031-20738-9_63.

10. Q. Yue, C. Zhong, and H. Lei, "Quantum Sealed-Bid Auction Protocol With Post-Confirmation Based on Blind Signature," *Quantum Information Processing* 23, no. 3 (2024): 70, https://doi.org/10.1007/s11128-024-04268-2.

11. J. Zhong, B. Liao, Y. Shi, and R. Lu, "A Quantum Partially Blind Signature Scheme Without Entanglement," *International Journal of Modern Physics B* 36, no. 21 (2022): 2250128, https://doi.org/10.1142/S0217979222501284.

12. K. Xing, A.-H. Yin, and Y.-Q. Xue, "A Quantum Blind Signature Scheme Based on Dense Coding for Non-Entangled States," *Chinese Physics B* 33, no. 6 (2024): 060309, https://doi.org/10.1088/1674-1056/ad2bed.

13. X. Feng, H. Wu, X. Zhou, and Y. Yao, "Quantum Blind Signature Scheme for Supply Chain Financial," *Quantum Information Processing* 22, no. 1 (2022): 5, https://doi.org/10.1007/s11128-022-03763-8.

14. D. J. Egger, R. G. Gutiérrez, J. C. Mestre, and S. Woerner, "Credit Risk Analysis Using Quantum Computers," *IEEE Transactions on Computers* 70, no. 12 (2020): 2136–2145, https://doi.org/10.1109/TC.2020.3038063.

15. P. Griffin and R. Sampat, "Quantum Computing for Supply Chain Finance," in *2021 IEEE International Conference on Services Computing (SCC)* (IEEE, 2021), 456–459, https://doi.org/10.1109/SCC53864.2021.00066.

16. F. Gao, S.-J. Qin, F.-Z. Guo, and Q.-Y. Wen, "Cryptanalysis of the Arbitrated Quantum Signature Protocols," *Physical Review A: Atomic, Molecular, and Optical Physics* 84, no. 2 (2011): 022344, https://doi.org/10.1103/PhysRevA.84.022344.

17. L. Ding, X. Xin, Q. Yang, and Y. Sang, "Security Analysis and Improvements of Xor Arbitrated Quantum Signature-Based Ghz State," *Modern Physics Letters A* 37, no. 02 (2022): 2250008, https://doi.org/10.1142/S0217732322500080.

18. B.-M. Zhou, L.-D. Lin, W. Wang, and Y. Liu, "Security Analysis of Particular Quantum Proxy Blind Signature Against the Forgery Attack," *International Journal of Theoretical Physics* 59, no. 2 (2020): 465–473, https://doi.org/10.1007/s10773-019-04340-z.

19. X.-Q. Cai, T.-Y. Wang, C.-Y. Wei, and F. Gao, "Cryptanalysis of Quantum Digital Signature for the Access Control of Sensitive Data," *Physica A: Statistical Mechanics and its Applications* 593 (2022): 126949, https://doi.org/10.1016/j.physa.2022.126949.

20. T. Shang, Q. Lei, and J. Liu, "Quantum Random Oracle Model for Quantum Digital Signature," *Physical Review A* 94, no. 4 (2016): 042314, https://doi.org/10.1103/PhysRevA.94.042314.

21. T. Shang, R. Chen, and Q. Lei, "Quantum Random Oracle Model for Quantum Public-Key Encryption," *IEEE Access* 7 (2019): 130 024–130 031, https://doi.org/10.1109/ACCESS.2019.2940406.

22. L. Yang, C. Xiang, and B. Li, "Quantum Probabilistic Encryption Scheme Based on Conjugate Coding," *China Communications* 10, no. 2 (2013): 19–26, https://doi.org/10.1109/CC.2013.6472855.

23. X. Xin, Q. Yang, and F. Li, "Quantum Proxy Signature With Provable Security," *Modern Physics Letters A* 35, no. 24 (2020): 2050197, https://doi.org/10.1142/S0217732320501977.

24. Y. Yeo and W. K. Chua, "Teleportation and Dense Coding With Genuine Multipartite Entanglement," *Physical Review Letters* 96, no. 6 (2006): 060502, https://doi.org/10.1103/PhysRevLett.96.060502.

25. P. O. Boykin and V. Roychowdhury, "Optimal Encryption of Quantum Bits," *Physical Review A* 67, no. 4 (2003): 042317, https://doi.org/10.1103/PhysRevA.67.042317.

26. F. He, X. Xin, C. Li, and F. Li, "Security Analysis of the Semi-Quantum Secret-Sharing Protocol of Specific Bits and Its Improvement," *Quantum Information Processing* 23, no. 2 (2024): 51, https://doi.org/10.1007/s11128-023-04255-z.

27. S. M. Barnett, A. Chefles, and I. Jex, "Comparison of Two Unknown Pure Quantum States," *Physics Letters A* 307, no. 4 (2003): 189–195, https://doi.org/10.1016/S0375-9601(02)01602-X.

28. S. Liu, Z. Wan, Y. Yuan, Q. Dong, B. Yang, and F. Hao, "Efficient Certificateless Blind Signature Scheme With Conditional Revocation for Mobile Crowdsensing Within Smart City," *IEEE Internet of Things Journal* 11, no. 9 (2024): 15 985–15 997, https://doi.org/10.1109/JIOT.2024.3350023.

29. S. Singh, S. Rawal, S. Padhye, and N. Tiwari, "Identity Based Proxy Blind Signature Scheme Using Ntru Lattices," *Information and Computation* 304 (2025): 105284, https://doi.org/10.1016/j.ic.2025.105284.

30. M. Bellare and P. Rogaway, "Random Oracles Are Practical: A Paradigm for Designing Efficient Protocols," in *Proceedings of the 1st ACM Conference on Computer and Communications Security*, CCS '93

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(New York, NY, USA: Association for Computing Machinery, 1993), 62–73, https://doi.org/10.1145/168588.168596.

31. R. P. Parameswarath, C. Wang, and B. Sikdar, "A Quantum Safe Mutual Authentication Protocol for Smart Meter Communications With Experimental Evaluation," *IEEE Transactions on Network Science and Engineering* (2024), https://doi.org/10.1109/TNSE.2024.3427110.

32. X. Xin, L. Ding, C. Li, Y. Sang, Q. Yang, and F. Li, "Quantum Public-Key Designated Verifier Signature," *Quantum Information Processing* 21, no. 1 (2022): 33, https://doi.org/10.1007/s11128-021-03387-4.

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